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AI-based automated grading in higher education: A scoping review and landscape analysis for sustainable education

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Abstract

The growing demand for scalable and high-quality education aligned with sustainable development goals increases pressure on assessment processes in higher education. Artificial intelligence (AI) has emerged as a potential solution for automated grading; however, its capabilities and limitations remain unclear. This study presents a scoping review of AI-based automated grading systems based on 72 studies retrieved from the Scopus database. The analysis maps AI techniques, assessment tasks, evaluation metrics, and dataset characteristics. The results show a shift toward large language models, which dominate recent research. AI systems demonstrate strong performance in structured tasks, such as short-answer grading, but remain limited in complex and subjective assessments. The findings also reveal challenges related to small datasets, lack of standardization, and concerns about fairness and reliability. Overall, the study provides a structured overview of current capabilities and gaps, highlighting the role of AI in supporting scalable and sustainable education systems.

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Keywords: Artificial intelligence; automated grading; higher education; large language models; sustainable education; scoping review

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1. Introduction

The increasing demand for scalable, high-quality, and sustainable education places growing pressure on higher education systems, particularly in the assessment of student performance. In the context of SDG 4, sustainable education requires not only broader access to learning, but also fair, timely, scalable, and pedagogically valid assessment under conditions of growing enrolment, limited instructional resources, and increasing demand for feedback. Traditional grading of open-ended tasks, such as essays and short answers, is time-consuming, difficult to scale, and often affected by inconsistencies across evaluators [1,2].

This challenge has become more complex with the rapid development of artificial intelligence (AI) and its increasing use by students. As learners rely on AI tools to generate, revise, or support their responses, submitted work becomes more variable and more difficult to evaluate through conventional manual approaches. At the same time, AI also offers new possibilities for supporting assessment processes by improving efficiency, consistency, and feedback speed.

AI-based automated grading has therefore emerged as a promising application of digital intelligence in higher education. Early approaches focused mainly on machine learning methods for automated essay scoring [1], while more recent studies have explored deep learning and large language models (LLMs) for evaluating textual responses [3,4]. These systems can potentially reduce instructor workload, support large-scale assessment, and provide faster feedback to students. However, their contribution to sustainable education depends on whether they preserve fairness, transparency, reliability, and meaningful human oversight.

Despite rapid progress, the effectiveness of AI-based grading systems remains inconsistent. Prior research highlights variability in agreement with human evaluators, sensitivity to task complexity, and limitations related to fairness, transparency, reproducibility, and reliability [3–5]. Moreover, it remains unclear which types of assessment tasks are currently well supported by AI, which AI approaches are most appropriate for different grading contexts, and where significant methodological and practical gaps persist.

To address this gap, this study presents a scoping review of AI-based automated grading in higher education. Based on 72 studies retrieved from the Scopus database, the paper maps AI techniques, assessment tasks, evaluation methods, dataset characteristics, and reported limitations. The findings aim to clarify the current capabilities and constraints of AI-based grading systems and assess their potential role in supporting scalable and sustainable higher education assessment.

To guide the analysis and align the study with the role of digital intelligence in sustainable development, the following research questions are defined:

RQ1. What is the current landscape of AI-based automated grading in higher education in terms of models, assessment tasks, and evaluation approaches?

RQ2. How effective are AI-based grading systems in supporting scalable and reliable assessment across different types of educational tasks?

RQ3. What are the key limitations, challenges, and research gaps that affect the adoption of AI-based grading systems in sustainable education environments?

2. Methodology

This study uses a scoping review design to map AI-based automated grading in higher education. This approach was selected because the field is rapidly evolving and methodologically diverse. The review was guided by PRISMA-ScR principles [6,7].

The search was conducted in Scopus on March 23, 2026, using title, abstract, and keyword fields. Search terms covered student assessment, automated grading, higher education, and AI, including “student assessment”, “grading”, “automated grading”, “higher education”, “universit*”, “artificial intelligence”, “machine learning”, “natural language processing”, “learning analytics”, “large language model*”, and “ChatGPT”. The search was limited to final journal articles, conference papers, and reviews from 2013 to 2026 available in Scopus by the search date.

Studies were included if they applied AI-based methods to automated grading or scoring of student responses or assignments and reported empirical, experimental, or system-evaluation results. Studies focused only on feedback generation, peer assessment, non-AI rule-based grading, or lacking sufficient methodological detail were excluded.

The selection process was conducted collaboratively by the authors and included title screening, abstract screening, availability checking, and full-text reading. The initial export contained 701 records; 300 remained after title screening, 130 after abstract screening, and 72 studies were included after full-text review.

Data were extracted into a structured coding table covering AI type, task type, domain, assessment type, automation level, human comparison, dataset size, metrics, key findings, limitations, year, source, funding, and document type. The analysis combined descriptive quantitative mapping and qualitative comparative synthesis to identify trends, methodological differences, limitations, and gaps.

ChatGPT was used only for language editing and paraphrasing. All search, screening, extraction, analysis, and conclusions were conducted independently by the authors.

3. Results

3.1. Overview of the Selected Studies

A total of 72 studies were included in the final sample. The studies cover the period from 2013 to 2026, with a strong increase in publication activity in recent years, especially after the wider adoption of deep learning and LLM-based approaches.

The sample includes both journal articles and conference papers. Journal articles represent the larger share (n = 43), while conference papers also form a substantial part of the literature (n = 29). The studies were published across diverse international venues, including Springer Nature, IEEE, MDPI, Taylor & Francis, ACM, and Elsevier, as well as specialized educational technology and domain-specific journals.

Affiliation data indicate that all included studies involved university or academic institutions, while several also included technology-oriented partners, research centers, or applied educational projects. The sample covers authors affiliated with institutions across 39 countries, with 16 studies involving multi-country affiliations. This confirms the interdisciplinary and international character of AI-based automated grading research.

Funding information was not consistently reported, which limits systematic interpretation of funding patterns. Overall, the literature remains fragmented across AI techniques, task types, datasets, and evaluation practices, supporting the use of a scoping review approach.

3.2. Landscape of AI-based Automated Grading

Table 1. Distribution of studies by year and AI category.

Year	ML	DL	LLM	Other	Total
2013	0	0	0	1	1
2017	0	0	0	1	1
2019	2	1	0	0	3
2020	1	1	0	0	2
2021	0	2	0	0	2
2022	2	0	0	0	2
2023	0	2	1	0	3
2024	1	2	9	0	12
2025	3	8	30	0	41
2026	0	1	5	0	6

The selected studies (n = 72) show a clear temporal shift in AI-based automated grading. Early work from 2013 to 2017 was limited and mainly exploratory, relying on non-standard or rule-based approaches. From 2019 to 2022, the

field became more active and was dominated by traditional machine learning and deep learning methods. Since 2023, large language models have rapidly entered the field, becoming the dominant approach in 2024–2025. Overall, the results indicate a transition from task-specific ML/DL models toward more flexible LLM-based grading systems, although this shift also raises new concerns about reliability, reproducibility, and validation.

Table 2. Distribution of assessment task types.

Task Type	Count	Percentage
Short Answer	32	44%
Essay	19	26%
Code	6	8%
Numerical	2	3%
Other / Mixed	13	18%

In terms of assessment tasks, the reviewed studies show an uneven distribution. Short-answer grading is the most common task type (n = 32, 44%), followed by essay assessment (n = 19, 26%). Code evaluation is less frequent (n = 6, 8%), while numerical problem grading is rarely addressed (n = 2, 3%). Mixed or less clearly defined tasks account for 13 studies (18%). This distribution suggests that AI-based grading is most mature in structured and semi-structured tasks with explicit criteria, whereas more complex, domain-specific, or authentic assessment formats remain underexplored.

3.3. Effectiveness of AI-based Grading Systems

Table 3. Distribution of dataset sizes across studies.

Dataset Size Category	Range (samples)	Count	Percentage
Small	< 500	29	40%
Medium	500 – 5,000	18	25%
Large	5,000 – 50,000	11	15%
Very Large	> 50,000	5	7%
Not Reported	—	9	13%

The analysis of dataset characteristics reveals an uneven distribution of dataset sizes across the selected studies (Table 3). A substantial proportion of studies relies on small datasets (n = 29, 40%), typically containing fewer than 500 samples and often limited to domain-specific or experimental settings.

Medium-sized datasets (n = 18, 25%) and large datasets (n = 11, 15%) are less frequently used, indicating that many models are not trained or evaluated on sufficiently diverse data. Very large datasets (n = 5, 7%) are relatively rare and are mainly associated with recent studies involving large language models.

Additionally, a notable number of studies (n = 9, 13%) do not explicitly report dataset size, which limits reproducibility and comparability across studies.

Overall, these findings suggest that the limited availability of large and diverse datasets remains a key constraint for developing robust, generalizable, and scalable AI-based grading systems.

Table 4. Evaluation metrics used in AI-based grading studies.

Metric Type	Examples	Count	Usage Frequency	Purpose
Accuracy-based	Accuracy, Precision, Recall	24	High	Classification performance
Error-based	RMSE, MAE	20	Medium	Regression / scoring error
Correlation-based	Pearson, Spearman	33	High	Alignment with human grades
Agreement-based	Cohen's Kappa	33	High	Human-AI agreement
Other	F1-score	14	Medium	Balanced evaluation

As shown in Table 4, the reviewed studies use diverse evaluation metrics. Correlation-based and agreement-based measures are the most frequently reported (n = 33 each), reflecting the importance of comparing AI-generated grades with human assessment. Accuracy-based metrics (n = 24), including accuracy, precision, and recall, are mainly used in classification-oriented tasks, while error-based metrics such as RMSE and MAE (n = 20) are used in regression or score-prediction settings. F1-score and related balanced classification measures appear in 14 studies. Overall, the frequent use of multiple metrics reflects the complexity of automated grading, but the absence of a shared evaluation framework limits comparability and weakens cumulative evidence in the field. Studies may report multiple evaluation metrics; therefore, counts are not mutually exclusive.

3.4. Limitations and Research Gaps

Despite rapid growth in AI-based automated grading research, several limitations remain evident. First, many studies rely on small or domain-specific datasets: 40% of the reviewed studies (n = 29) used fewer than 500 samples. This limits model generalizability and reduces confidence in applying these systems across disciplines, institutions, and student populations.

Second, current evidence remains strongly task dependent. Short-answer grading dominates the sample (44%), while essays (26%), programming assignments (8%), and numerical problem solving (3%) are less represented. This suggests that AI-based grading is better validated for structured or semi-structured tasks than for open-ended, creative, multimodal, or highly domain-specific assessments.

Third, grading outcomes are highly sensitive to evaluation design. Rubric formulation, scoring criteria, dataset selection, model configuration, and, in LLM-based systems, prompt design can substantially affect results. This raises concerns about consistency and reproducibility across studies.

A further limitation is the lack of standardized evaluation methods. The reviewed studies use diverse metrics, including accuracy-based measures (n = 24), correlation-based measures (n = 33), and agreement-based measures (n = 33), which makes direct comparison difficult. Moreover, high performance in one dataset does not necessarily demonstrate educational validity across contexts.

Finally, fairness, bias, and transparency remain insufficiently addressed. AI systems may grade differently depending on language use, writing style, student background, or task framing. This is critical for sustainable education, where scalable assessment must remain fair, explainable, and pedagogically valid.

Overall, AI-based automated grading shows strong potential, but reliable adoption in large-scale higher education requires larger and more diverse datasets, stronger validation protocols, standardized metrics, fairness testing, and continued human oversight.

4. Discussion

The findings show a clear shift from traditional ML and DL approaches toward LLM-based automated grading. However, this shift is not a simple technological replacement. ML remains useful for structured and interpretable grading, while DL improves semantic representation but requires larger datasets and offers limited transparency. LLMs provide greater flexibility for open-ended responses and feedback generation, but raise concerns about prompt sensitivity, reproducibility, model-version instability, opacity, and scoring inconsistency. The key trade-off is therefore between semantic flexibility and assessment reliability. Thus, ML/DL approaches appear more defensible where scoring criteria are stable and datasets are task-specific, whereas LLM-based grading is more adaptable but requires stronger validation and governance.

The results also show that automated grading is strongly task dependent. Current evidence is strongest for short-answer and other structured or semi-structured tasks, where expected answers, rubrics, and scoring criteria are explicit [8]. More complex formats, including essays, programming, numerical reasoning, project-based work, oral assessment, creative tasks, and multimodal assignments, remain less consistently validated. Thus, AI-based grading should be used primarily as a supportive tool rather than a universal substitute for human assessment.

Another important issue is validation quality. Many studies use small or domain-specific datasets, and evaluation metrics vary substantially, limiting comparability across studies. High accuracy in one dataset does not necessarily

indicate educational validity across disciplines, languages, institutions, or student populations. For LLM-based grading, this problem is amplified by prompt design, rubric clarity, model configuration, and version changes.

From a sustainable development perspective, AI-based grading can support SDG 4 by improving scalability, reducing instructor workload, and enabling faster feedback. However, this contribution depends on fairness, transparency, reliability, pedagogical alignment, and meaningful human oversight. If poorly validated or applied only to easily measurable tasks, AI grading may reinforce inequality and weaken assessment quality.

Overall, AI-based grading is most promising as an assistive component of assessment systems. Future research should focus on cross-task validation, standardized evaluation protocols, fairness testing, explainability, and complex learning outcomes that cannot be easily reduced to automated scores.

5. Conclusion

This scoping review analyzed 72 Scopus-indexed studies on AI-based automated grading in higher education. The findings show a clear shift from ML/DL approaches toward LLM-based grading, but this shift does not mean that AI is ready to replace human judgment across all assessment contexts.

AI-based grading is most mature in structured and semi-structured tasks, especially short-answer grading, while complex formats such as essays, programming, numerical reasoning, creative work, project-based assessment, and multimodal tasks remain less consistently validated.

The review identifies persistent challenges related to small datasets, fragmented metrics, limited standardization, fairness, transparency, reliability, and reproducibility. From a sustainable development perspective, AI grading can support SDG 4 by improving scalability, reducing instructor workload, and enabling faster feedback, but only with human oversight and robust validation. Future research should focus on larger datasets, standardized evaluation protocols, fairness testing, and real-world validation across disciplines.

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References

[1] Shermis, M., Burstein, J., 2013. Handbook of Automated Essay Evaluation: Current Applications and New Directions. Routledge.
[2] Williamson, D.M., Xi, X., Breyer, F.J., 2012. A framework for evaluation and use of automated scoring. Educational Measurement: Issues and Practice 31(1), 2–13. https://doi.org/10.1111/j.1745-3992.2011.00223.x
[3] Zhai X. 2023. ChatGPT for Next Generation Science Learning. XRDS 29, 3 (Spring 2023), 42–46. https://doi.org/10.1145/3589649
[4] Kasneci, E., Sessler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F. et al. (2023). Chatgpt for Good? on Opportunities and Challenges of Large Language Models for Education. *Learning and Individual Differences, 103,* Article ID: 102274. https://doi.org/10.1016/j.lindif.2023.102274
[5] Bubeck, S., Chandrasekaran, V., Eldan, R., Gehrke, J., Horvitz, E., Kamar, E., Lee, P., Lee, Y.T., Li, Y., Lundberg, S., Nori, H., Palangi, H., Ribeiro, M.T., Zhang, Y., 2023. Sparks of Artificial General Intelligence: Early experiments with GPT-4. arXiv preprint arXiv:2303.12712.
[6] Arksey, H., O'Malley, L., 2005. Scoping studies: towards a methodological framework. International Journal of Social Research Methodology 8(1), 19–32. https://doi.org/10.1080/1364557032000119616
[7] Tricco, A.C., Lillie, E., Zarin, W., O'Brien, K.K., Colquhoun, H., Levac, D., Moher, D., Peters, M.D.J., Horsley, T., Weeks, L., Hempel, S., Akl, E.A., Chang, C., McGowan, J., Stewart, L., Hartling, L., Aldcroft, A., Wilson, M.G., Garritty, C., Lewin, S., Godfrey, C.M., Macdonald, M.T., Langlois, E.V., Soares-Weiser, K., Moriarty, J., Clifford, T., Tunçalp, Ö., Straus, S.E., 2018. PRISMA Extension for Scoping Reviews (PRISMA-ScR): Checklist and Explanation. Annals of Internal Medicine 169(7), 467–473. https://doi.org/10.7326/M18-0850
[8] Burrows, S., Gurevych, I., Stein, B., 2015. The eras and trends of automatic short answer grading. International Journal of Artificial Intelligence in Education 25(1), 60–117. https://doi.org/10.1007/s40593-014-0026-8